



Smart NGO Donation and Volunteer Management Platform

CSA0905 – Programming In Java

**Guided by:
Dr. Sivasankar C**

**Presented By:
Y.Parveen(192325150)
G.Deevena(192311391)
K.Ramya(192311149)**

Review Feedback

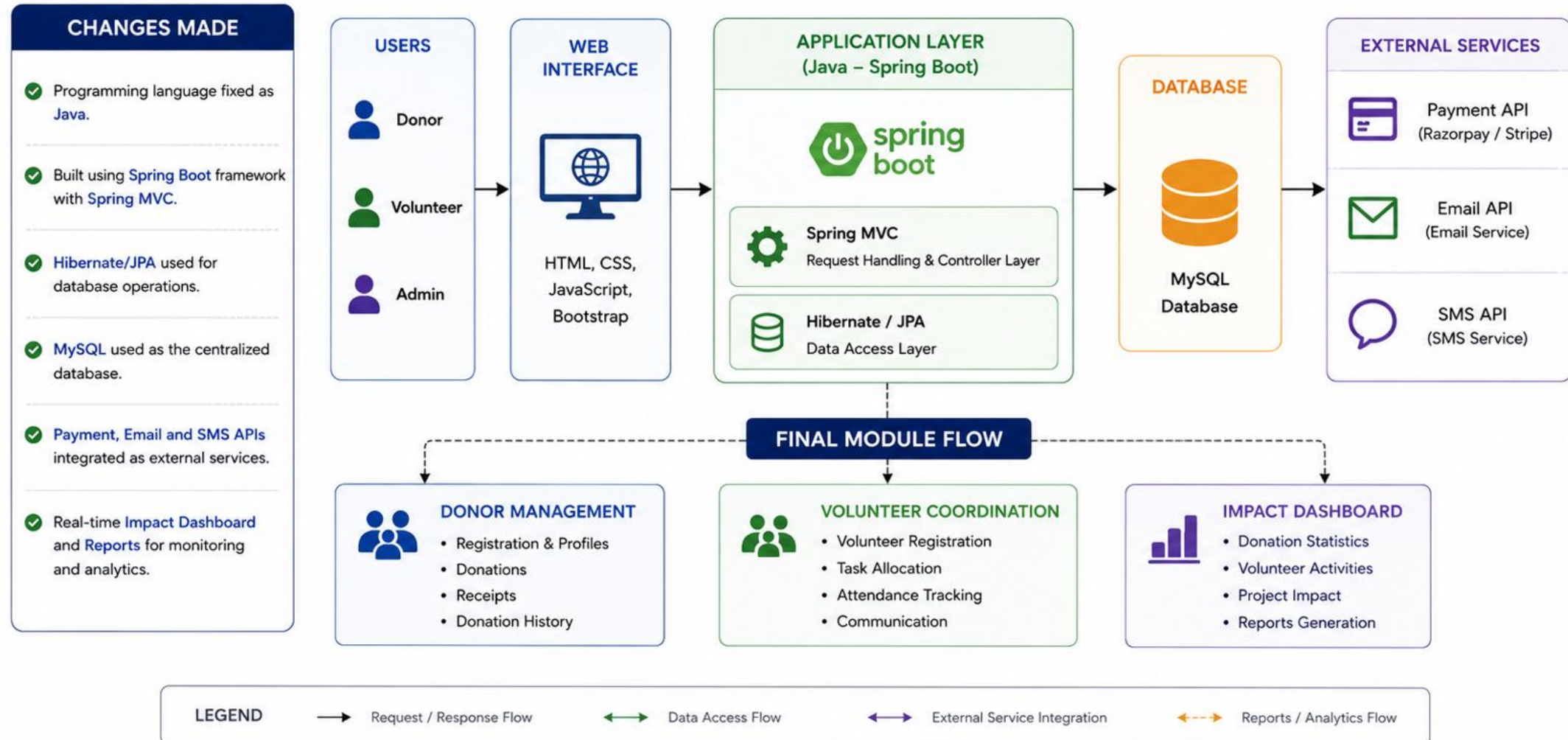
Feedback Received	Corrective Actions
Clarify the project objectives and scope	Refined objectives to be specific and measurable
Strengthen the literature survey	Added recent research papers and identified the research gap
Improve comparison with existing systems	Added detailed Existing vs Proposed comparison
Clarify system architecture and workflow	Added system architecture, workflow, and module descriptions
Specify technical requirements	Defined functional, non-functional, software, database, and API requirements
Improve project planning	Added WBS, timeline, milestones, and risk analysis



Engineer to Excel



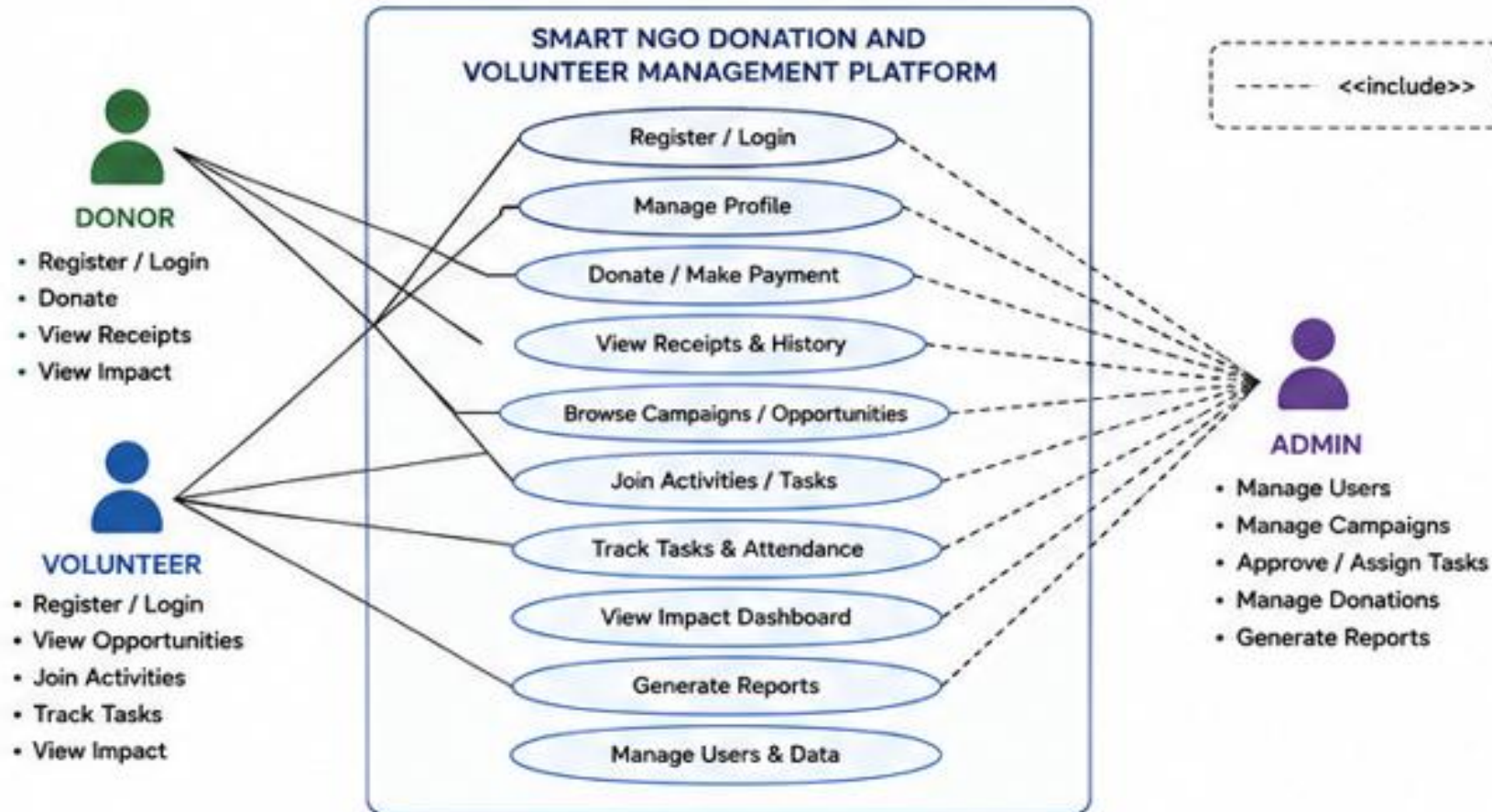
Updated Architecture



Module Design

Module	Purpose	Data Flow	Interface
Donor Management	Manage donors and donations	Donor → Donation → Database	Donor Registration, Donation, History
Volunteer Coordination	Manage volunteers and activities	Volunteer → Task/Event → Database	Registration, Tasks, Attendance
Impact Dashboard	Monitor NGO performance	Database → Analytics → Dashboard	Charts, Reports, Impact Metrics

Use Case Diagram

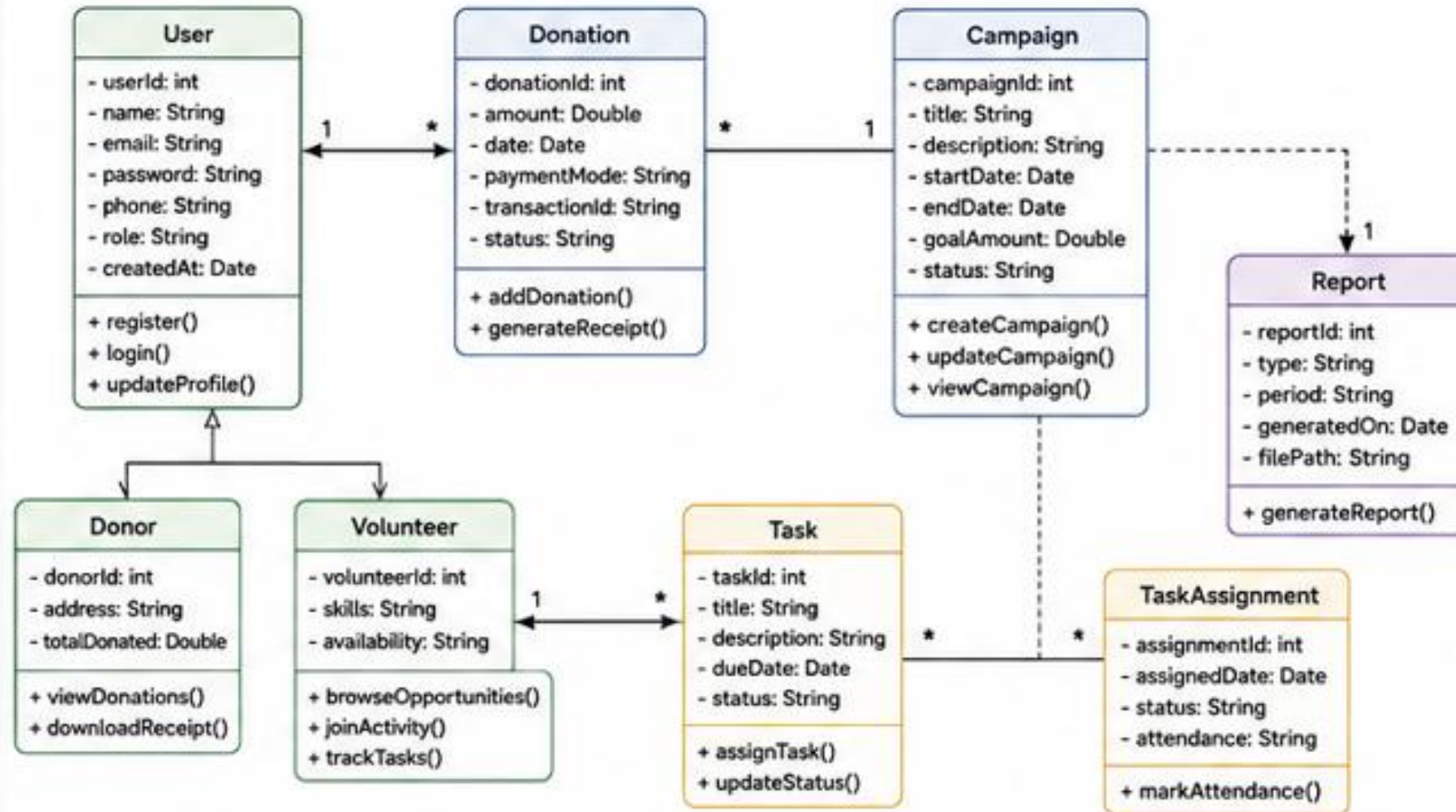




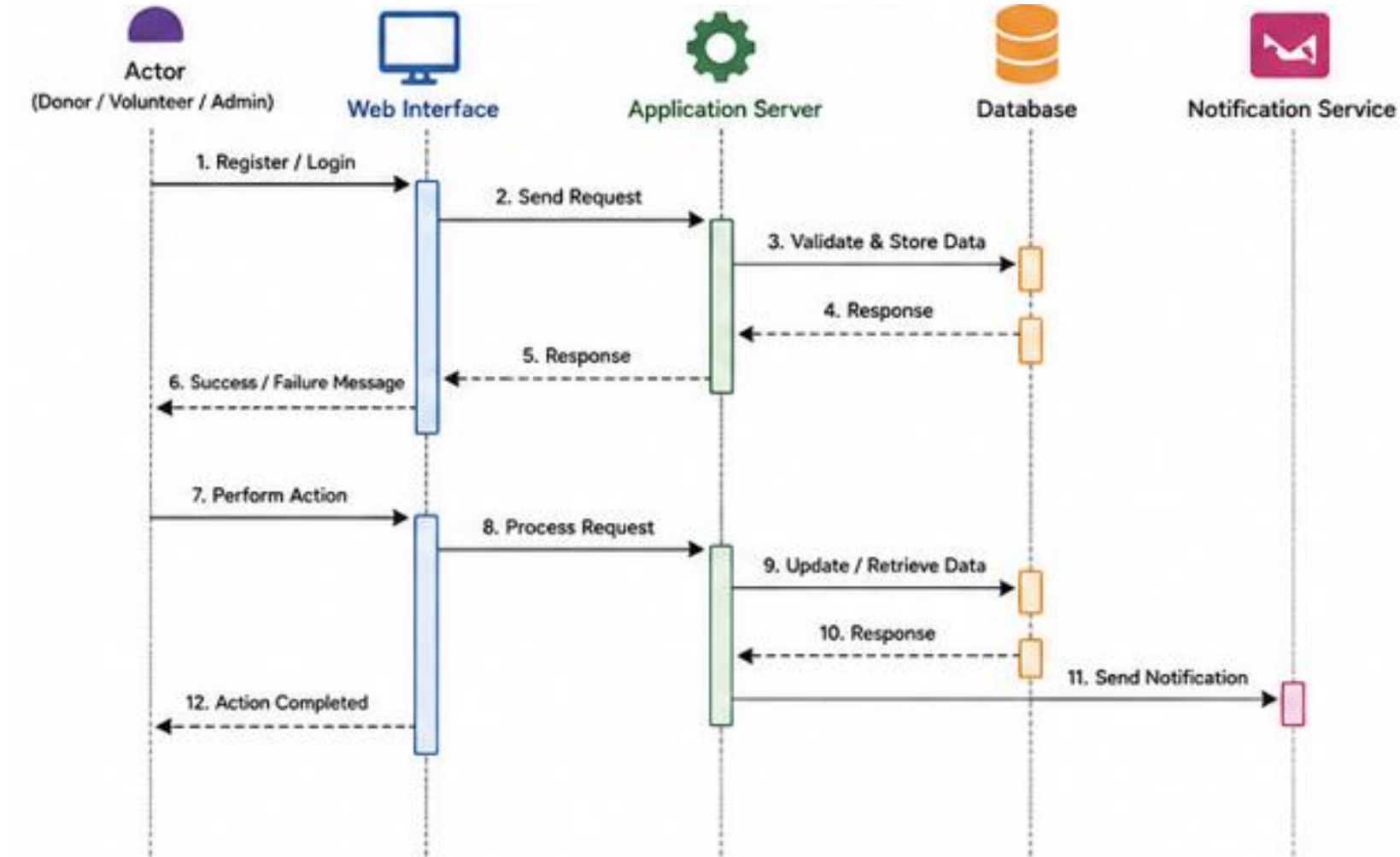
Engineer to Excel



Class Diagram



Sequence Diagram

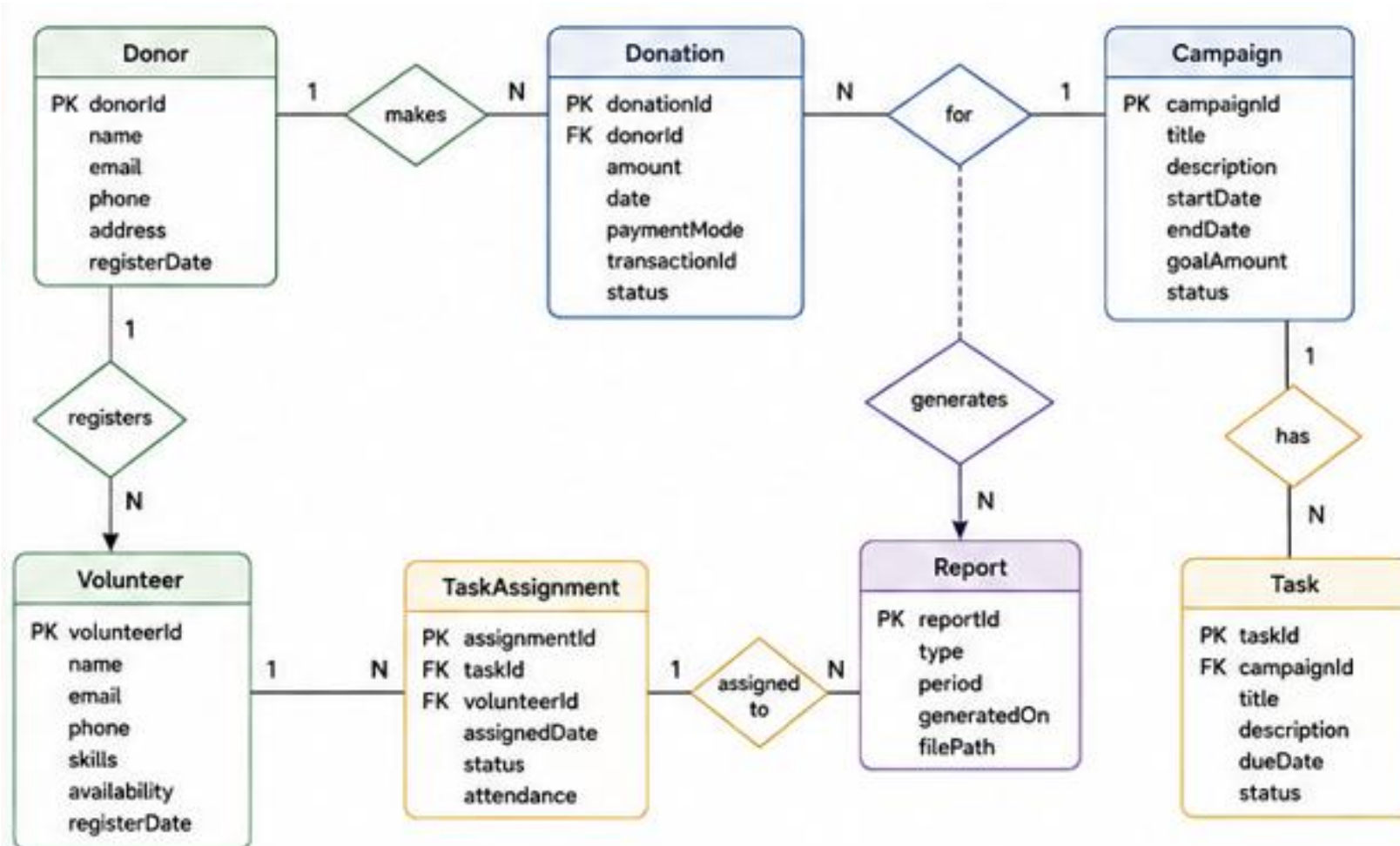




Engineer to Excel



ER Diagram





Technology Stack



- **Programming Language:** Java
- **Frameworks:** Spring Boot, Spring MVC
- **Libraries:** Hibernate/JPA, Bootstrap, Chart.js
- **Database:** MySQL
- **Development Tools:** IntelliJ IDEA / Eclipse, Git & GitHub
- **APIs:** Razorpay/Stripe Payment API, Email/SMS Notification API
- **Hardware:** Not Applicable – Web-based Application



Implementation



Completion Status

- **Overall Completion:** 75 %
- **Frontend:** Web interface and user screens developed.
- **Backend:** Java + Spring Boot application structure implemented.
- **Database:** MySQL database and entity integration implemented.
- **Core Modules:** Donor Management, Volunteer Coordination, and Impact Dashboard under development/implementation.

Code Summary

- **Java + Spring Boot** used for application development.
- **Spring MVC** handles requests and application flow.
- **Hibernate/JPA** manages database operations.
- **MySQL** stores donor, volunteer, donation, and activity data.



Engineer to Excel

Screenshots



1. Login / Registration Page


Smart NGO
Together We Can Make a Difference

Welcome Back!

Login to your account

Email

Password

☐ Remember me [Forgot Password?](#)

Login

Don't have an account? [Register here](#)

Create an Account

Register to get started

Full Name

Email

Phone

Password

Confirm Password

Role

Select Role

Register

Already have an account? [Login here](#)

2. Donor Management Page

Smart NGO
Dashboard
Donors
Donations
Receipts
Campaigns
Reports
Messages
Settings

Donor Management
+ Add Donor
Export

Total Donors
245

Active Donors
198

Total Donations
₹ 1,25,000

This Month
₹ 25,000

Donor List

ID	Name	Email	Phone	Total Donations (₹)	Last Donation	Status	Actions
D001	Rahul Patil	rahul@gmail.com	9876543210	15,000	01 May 2024	Active	Edit Delete
D002	Priya Sharma	priya@gmail.com	8765432109	10,500	28 Apr 2024	Active	Edit Delete
D003	Amit Verma	amit@gmail.com	9123456780	7,500	25 Apr 2024	Active	Edit Delete
D004	Neha Joshi	neha@gmail.com	9988776655	5,000	20 Apr 2024	Inactive	Edit Delete

3. Volunteer Coordination Page

Smart NGO
Dashboard
Donors
Volunteers
Tasks
Attendance
Events
Messages
Reports
Settings

Volunteer Management
+ Add Volunteer
Export

Total Volunteers
132

Active Volunteers
112

Tasks Assigned
45

Hours Contributed
320

Volunteer List

ID	Name	Email	Skills	Status	Tasks	Joined On	Actions
V001	Sneha Desai	sneha@gmail.com	Teaching	Active	5	10 Jan 2024	Edit Delete
V002	Rohan Kulkarni	rohan@gmail.com	Logistics	Active	3	12 Jan 2024	Edit Delete
V003	Pooja Mehta	pooja@gmail.com	Content Writing	Active	4	15 Jan 2024	Edit Delete
V004	Karan Singh	karan@gmail.com	Event Mgmt	Inactive	1	18 Jan 2024	Edit Delete

4. Impact Dashboard

Smart NGO
Dashboard
Donors
Volunteers
Donations
Campaigns
Reports
Messages
Settings

Impact Dashboard
01 May 2024 - 31 May 2024
Admin

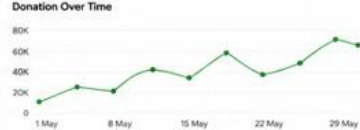
Total Donations
₹ 1,25,000
↑ 12.5% from last month

Total Donors
245
↑ 10.8% from last month

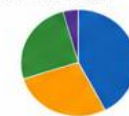
Total Volunteers
132
↑ 15.4% from last month

Projects Supported
18
↑ 8.6% from last month

Donation Over Time



Donations by Category



Recent Activities

- New donation of ₹ 5,000 received from Rahul Patil 2 hours ago
- Food Donation Drive completed 5 hours ago
- Tree Plantation event organized 1 day ago

Top Campaigns

Education for All	₹ 40,000 / ₹ 60,000	67%
Health & Wellness	₹ 30,000 / ₹ 50,000	60%
Clean Environment	₹ 20,000 / ₹ 40,000	50%



Engineer to Excel

UI



1. LOGIN



Welcome Back!

Login to your account

Email

Password

Role

Select Role

☐ Remember me

[Forgot Password?](#)

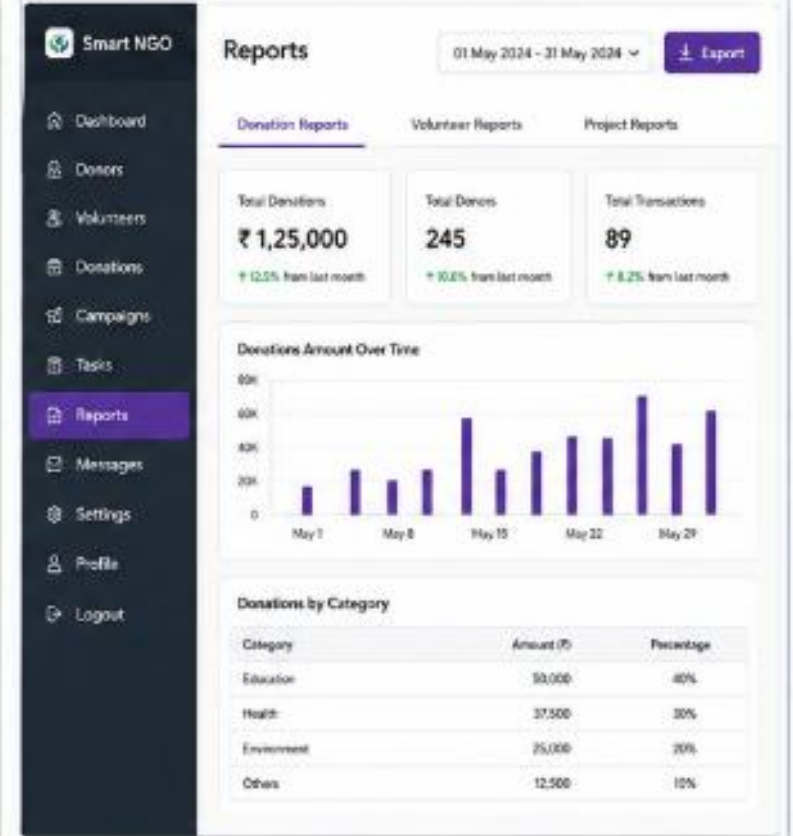
Login

Don't have an account? [Sign up](#)

2. DASHBOARD



3. REPORTS



Testing

Unit Tests

- Tested individual functions of Donor Management, Volunteer Coordination, and Impact Dashboard.
- Verified user registration, donation entry, volunteer task allocation, and data validation.
- Checked database operations using Hibernate/JPA and MySQL.

Integration Tests

- Tested interaction between Frontend → Java/Spring Boot → MySQL.
- Verified integration of modules and external APIs such as Payment and Email/SMS services.
- Tested end-to-end workflows such as registration → donation → database update → dashboard display.

Outputs

- Successful test cases displayed with Pass/Fail status.
- Verified correct data storage, retrieval, and dashboard updates.
- Identified and fixed errors during module and integration testing.

Challenges

Issues

- Managing multiple modules with a centralized database.
- Maintaining secure donor and volunteer information.
- Integrating payment and notification APIs.
- Ensuring accurate real-time dashboard data.

Solutions

- Used **Java + Spring Boot** with a modular architecture.
- Implemented **Hibernate/JPA + MySQL** for structured data management.
- Applied validation and secure authentication mechanisms.
- Tested module integration and data flow systematically.

Next Plan

- Complete remaining module implementation.
- Integrate and test external APIs.
- Perform final system and security testing.
- Complete deployment, documentation, and final presentation.

*Thank
you!*